

PURIX X1 · XRF gold analyzer

Karat isn't guessed. It's measured.

PURIX X1 measures up to 20 elements in gold in 30–60 seconds, with $\pm 0.05\%$ accuracy and without damaging the piece. No cutting, no touchstone, no acid.



Accuracy

$\pm 0.05\%$

RSD $\leq 0.05\%$ (Au $\geq 90\%$)

Measurement time

30–60 s

Per piece

Analysis range

1 ppm – 99.99 %

Au, Ag, Pt, Pd, Rh and
15 more elements

Effect on the piece

None

No cutting, scraping or acid

Measuring in four steps

It's that simple.

Karat, fineness and 20 elements – all with a single tap.

1

Place

Put the piece on the measuring window in the chamber. The camera shows the measuring spot on screen.

2

Start

Close the lid and tap START. The X-ray tube will not run until the lid is closed.

3

X-ray and fluorescence

The tube sends X-rays into the piece from below. Each element answers at its own energy: gold at 9.71 keV, silver at 22.16 keV.

4

Result

The detector counts these rays and the software turns them into ratios. Karat, fineness and all 20 elements appear on screen in 30–60 seconds.

Why XRF?

Criterion	Touchstone	Acid test	PURIX XRF
Effect on the piece	Leaves a scratch	Etches the surface	None
Result	Karat by eye	Karat range (e.g. 14–18)	Percentage of all 20 elements
Time	1–2 minutes	2–5 minutes	30–60 seconds
Plating detection	No	Limited	Reveals thin plating
Record	None	None	Saved measurement report

Specifications

Working principle
Energy dispersive XRF (EDXRF)

Detector
Si-PIN

Accuracy
±0.05 %

Repeatability (RSD)
≤ 0.05 % (Au ≥ 90 %)

Analysis range
1 ppm – 99.99 %

Detection limit
~2–5 ppm

Measurement time
30–60 s

Display
12.1" touchscreen

Sample camera
Full colour

Power supply
110 / 220 / 240 V

IP rating
IP55

Warranty
1 year

Measurable elements

Au, Ag, Pt, Pd, Rh, Ir, Ru, Cu, Zn, Ni, Cr, Co, Fe, Sn, Sb, Cd, Pb, Mn, Ga, In

All values are measured under the manufacturer's test conditions and may vary with the shape and surface of the sample.



Body

Porcelain-white body, graphite-black lid, champagne-gold display frame

Lid

Brushed gold handle; gas struts keep it open and let it close quietly

Ventilation and status

Perforated side ventilation grille; vertical gold status light beside the display

Who it is for

Jewellers

Measure karat in front of the customer when buying scrap, and back your price with numbers.

Pawnbrokers and gold lenders

File a measurement report for every piece taken as collateral; spot thin plating at the counter.

Bullion and coin dealers

Check the surface composition of bars and investment coins in seconds, without damage.

Refineries

Verify incoming material and outgoing bars on the same instrument, and track impurities.

Wholesalers and workshops

Verify alloy recipes before and after production; measure palladium in white gold.

Assay offices and laboratories

Check karat, verify hallmarks, detect impurities at ppm level and issue a measurement report.

Service and support

Installation & training

Installation, staff training and first calibration support; scope stated in the quote.

Licensing support

Support in preparing the technical documents for radiation licensing or registration.

1-year warranty

1-year manufacturer warranty; service support after the warranty.

Reference sample

A reference calibration sample is supplied; additional karat sets on request.

Radiation safety The tube shuts off the moment the lid opens. A dose-rate measurement report and the technical file for licensing are available on request.

Request a quote or an on-site demo

Scan the QR code or fill in the quote form on our website; we reply within one business day.

Web
www.purixxrf.com

Quote form
purixxrf.com/#teklif

Website languages
14

